EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

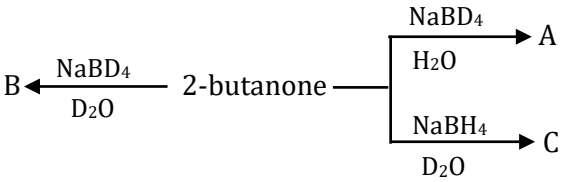
Metal hydride reduction ($LiAlH_4$; $NaBH_4$; DIBAL-H; B_2H_6; $LiBH_4$; $NaBH_3CN$)

1. The product of the reaction $Ph_2C = O \xrightarrow[H_3O^{\oplus}]{LiAlD_4}$ is

(A) $Ph_2CD(OH)$ (B) $Ph_2CH(OD)$ (C) $Ph_2CD(OD)$ (D) None

CRD086

2. Consider reduction of 2- butanone.



A, B and C are respectively

(A) $CH_3\underset{\underset{OH}{|}}{C}HCH_2CH_3$ in all cases

(B) $CH_3\underset{\underset{D}{|}}{\overset{\overset{OH}{|}}{C}}CH_2CH_3$, $CH_3\underset{\underset{D}{|}}{\overset{\overset{OD}{|}}{C}}CH_2CH_3$, $CH_3\underset{\underset{H}{|}}{\overset{\overset{OD}{|}}{C}}CH_2CH_3$

(C) $CH_3\underset{\underset{D}{|}}{\overset{\overset{OH}{|}}{C}}CH_2CH_3$, $CH_3\underset{\underset{H}{|}}{\overset{\overset{OH}{|}}{C}}CH_2CH_3$, $CH_3\underset{\underset{H}{|}}{\overset{\overset{OD}{|}}{C}}CH_2CH_3$ (D) $CH_3\underset{\underset{D}{|}}{\overset{\overset{OH}{|}}{C}}CH_2CH_3$, in all cases

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CRD087

Catalytic hydrogenation (Except alkene and alkyne)

3. The relative rates of hydrogenation is in the order of :

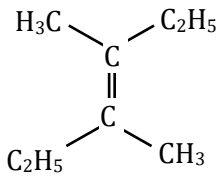
(A) $CH_2 = CH_2 > RCH = CH_2 > RCH = CHR > R_2C = CHR$

(B) $R_2C = CHR > RCH = CHR > RCH = CH_2 > CH_2 = CH_2$

(C) $RCH = CHR > R_2C = CHR > RCH = CH_2 > CH_2 = CH_2$

(D) $R_2C = CHR > CH_2 = CH_2 > RCH = CHR > RCH = CH_2$

CRD053

4. 

$\xrightarrow{H_2/Pt}$ x, x is

(A) Only (R,R) (B) Only (S,S) (C) Meso compound (D) Racemic mixture

CRD089

5. Which of the following reactions involves homogeneous reduction

(A) $CH_2 = CH_2 \xrightarrow{H_2/Ni} CH_3{-}CH_3$

(B) $CH_3{-}C \equiv C{-}CH_3 \xrightarrow{\text{Wilkinson's catalyst}} CH_3{-}CH{=}CH{-}CH_3$

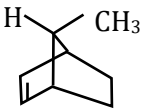
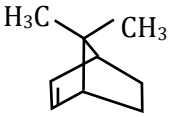
(C) $CH_3COCl \xrightarrow{H_2/Pd/CaCO_3/\text{quinoline}} CH_3CHO$

(D) $CH_3C \equiv CH \xrightarrow{H_2/Pd/CaCO_3/\text{quinoline}} CH_3CH = CH_2$

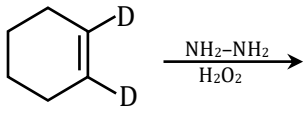
CRD088

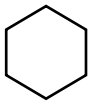
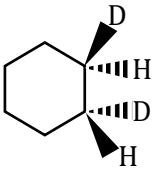
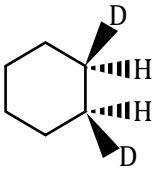
For More Join: @IITJEE_Advanced

6. Which undergoes catalytic hydrogenation much faster ?

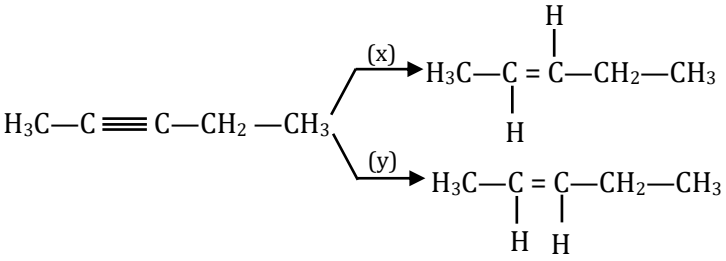
(A)  (B)  (C) equally (D) cannot be predicted

CRD090

7. 

(A)  (B)  (C)  (D) Both (B) and (C)

CRD003

8. 

In the above reaction the using reagents X and Y are

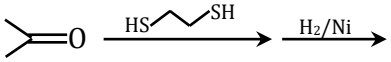
(A) Na/liq. NH_3 for X (B) H_2, Pd/$BaSO_4$ for Y

(C) BH_3–THF + CH_3 COOH for Y (D) All of these are correct

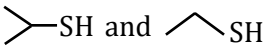
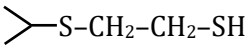
CRD091

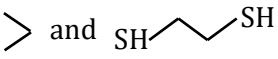
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Mozingo Reduction ($HSCH_2CH_2SH$ + Dry HCl + Raney Ni)

9. 

The end products of the reaction are :

(A)  (B) 

(C)  and H_2O (D) 

CRD001

Stephen's reduction ($SnCl_2$/HCl)

10. Stephen reduction ($SnCl_2$ / HCl) converts cyanides to

(A) Aldehydes (B) Ketones (C) Amines (D) Acids

CRD056

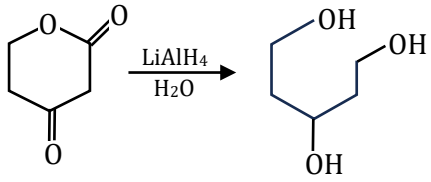
For More Join: @IITJEE_Advanced

EXERCISE (O-2)

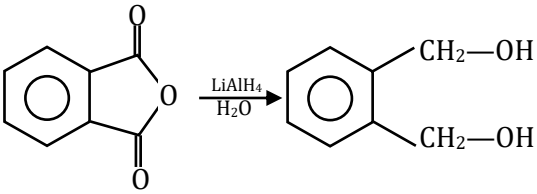
MULTIPLE CORRECT TYPE QUESTIONS

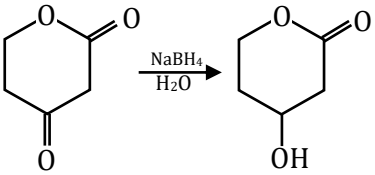
Metal hydride reduction ($LiAlH_4$; $NaBH_4$; DIBAL-H; B_2H_6; $LiBH_4$; $NaBH_3CN$)

1. Which of the following reactions is/ are correct?

(A) 

(B) $CH_3—C\equiv N \xrightarrow[H_2O]{LiAlH_4} CH_3—CH_2—NH_2$

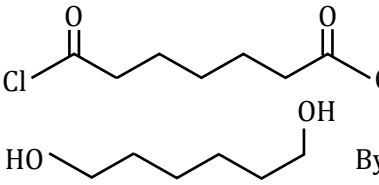
(C) 

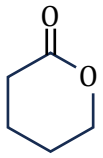
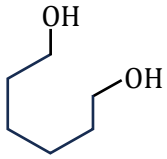
(D) 

CRD092

2. Which of the following is (are) not correct?

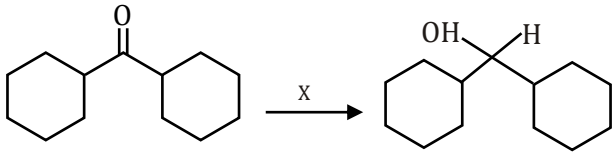
(A)  By LAH in ether, HI and red P.

(B)  OH reduced to By $NaBH_4$.

(C)  are reduced  by $NaBH_4$

(D) Oximes are reduced to 1° amines by LAH, H_2/Ni

CRD129

3. 

X is / are

(A) $NaBH_4$ / EtOH

(B) $LiAlH_4$ /THF

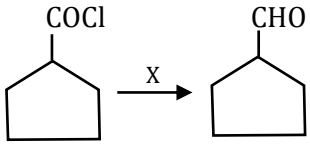
(C) Al $(O–iPr)_3$ / $CH_3—\underset{\underset{OH}{|}}{CH}—CH_3$

(D) Red P + HI

CRD094

For More Join: @IITJEE_Advanced

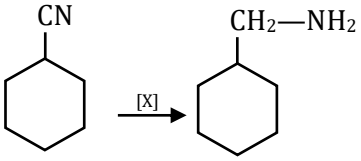
Catalytic hydrogenation (Except alkene and alkyne)

4. 

. X cannot be

(A) $NaBH_4$ /EtOH (B) $LiAlH_4$ / THF (C) Na/EtOH (D) H_2/Pd– $BaSO_4$

CRD095

5. 

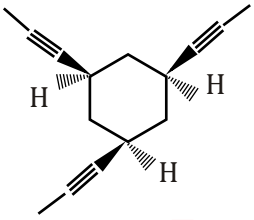
X can be

(A) H_2/Ni (B) $LiAlH_4$ (C) $NaBH_4$ (D) Na + EtOH

CRD096

COMPREHENSION TYPE QUESTIONS

Catalytic hydrogenation (Except alkene and alkyne)

6. 

(P)

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(P) $\xrightarrow{Na/NH_3(l)}$ Product(s)

The products (s) has/ have degree of unsaturation and number of isomeric product (s) formed are, respectively

(A) 1,4 (B) 4,2 (C) 4,1 (D) 2,4

CRD098

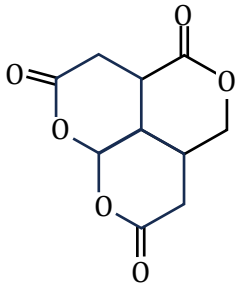
For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Metal hydride reduction ($LiAlH_4$; $NaBH_4$; DIBAL-H; B_2H_6; $LiBH_4$; $NaBH_3CN$)

1.  $\xrightarrow[2.LiAlH_4]{1.H_3O^+}$ A,

How many –OH group in the product (A) is

CRD101

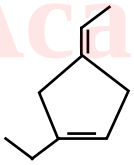
Catalytic hydrogenation (Except alkene and alkyne)

2. On catalytic reduction with H_2/Pt how many possible alkenes will give n-butane ?

CRD004

3. If the following compound is treated with Pd in excess of H_2 gas, how many stereoisomers of the product will be obtained ?

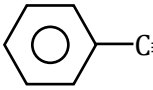
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CRD011

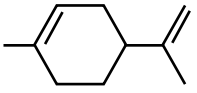
Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

4.  $\xrightarrow[-78^\circ C]{DiBAL-H}$ (X) if molecular weight of X is M then write your answer as M/100.

CRD130

5. For the product (P) given below, x is degree of unsaturation and y is total number of products then write your answer as sum of x + y is.

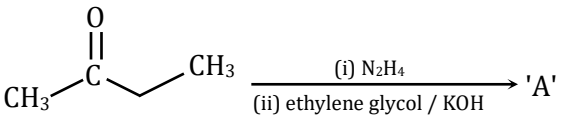
 $\xrightarrow[1\text{ eq.}]{H_2/Pt}$ P

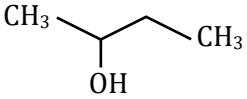
CRD131

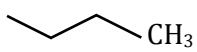
For More Join: @IITJEE_Advanced

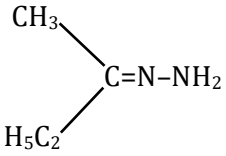
EXERCISE - JEE (Main) PYQ

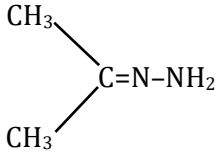
1. Identify 'A' in the following reaction : **[JEE (Main) 2024]**



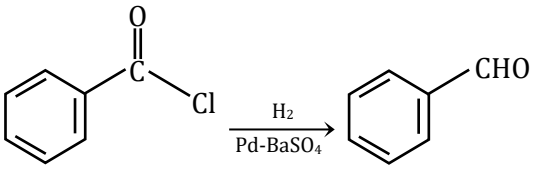
(1) 

(2) 

(3) 

(4) 

CRD132

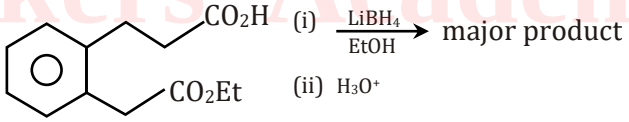
2. 

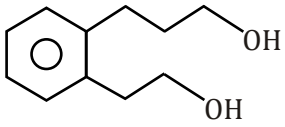
This reduction reaction is known as: **[JEE (Main) 2024]**

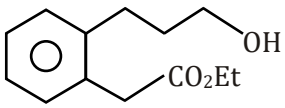
(1) Rosenmund reduction
(2) Wolff-Kishner reduction
(3) Stephen reduction
(4) Etard reduction

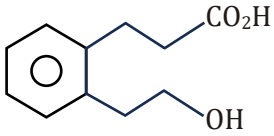
CRD133

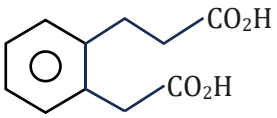
3. The major product formed in the following reaction is: **[JEE (Main) 2023]**



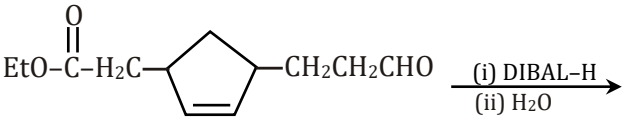
(1) 

(2) 

(3) 

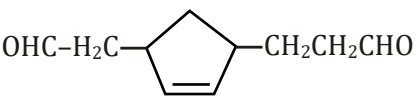
(4) 

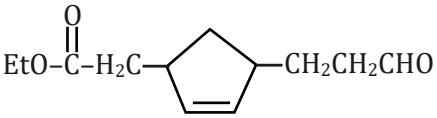
CRD111

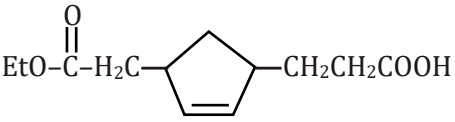
4. 

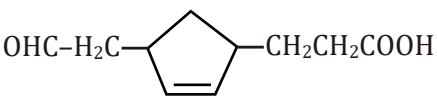
[JEE (Main) 2022]

Consider the above reaction and predict the major product.

(1) 

(2) 

(3) 

(4) 

CRD049

For More Join: @IITJEE_Advanced

5. **Statement I :** Sodium hydride can be used as an oxidising agent. **[JEE (Main) 2021]**

Statement II : The lone pair of electrons on nitrogen in pyridine makes it basic.

Choose the CORRECT answer from the options given below :

(1) Both statement I and statement II are false

(2) Statement I is true, but statement II is false

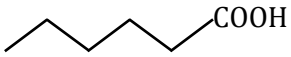
(3) Statement I is false, but statement II is true

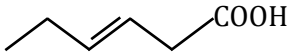
(4) Both statement I and statement II are true

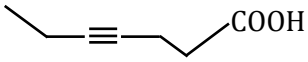
CRD045

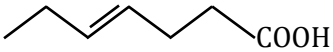
6. What is the product of following reaction ? **[JEE (Main) 2020]**

Hex-3-ynal $\xrightarrow[\text{(ii) } PBr_3 \\ \text{(iii) Mg/ether} \\ \text{(iv) } CO_2/H_3O^{\oplus}]{\text{(i) } NaBH_4}$?

(1) 

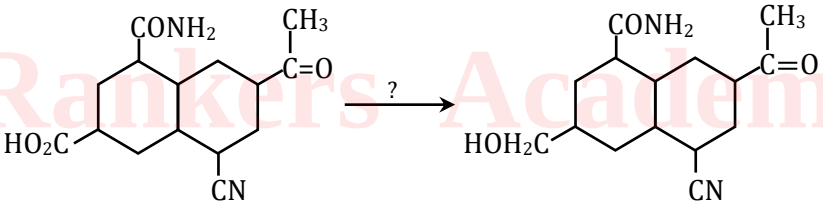
(2) 

(3) 

(4) 

CRD043

7. The most suitable reagent for the given conversion is : **[JEE (Main) 2020]**



(1) $LiAlH_4$

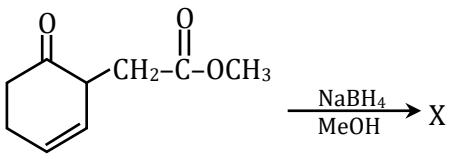
(2) $NaBH_4$

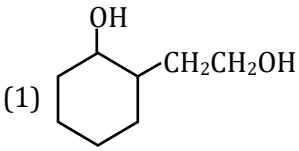
(3) H_2/Pd

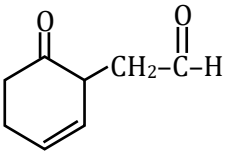
(4) B_2H_6

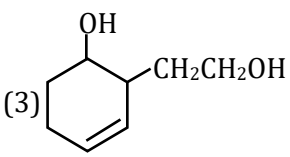
CRD044

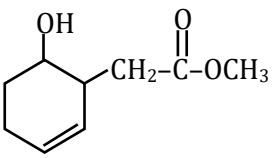
8. The major product 'X' formed in the following reaction is : **[JEE (Main) 2019]**



(1) 

(2) 

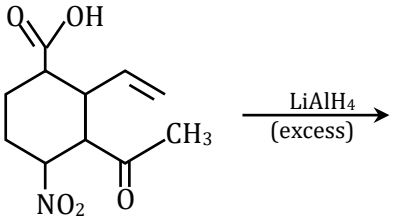
(3) 

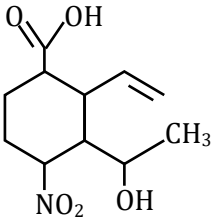
(4) 

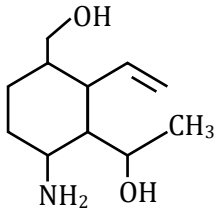
CRD037

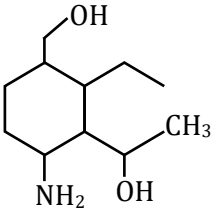
For More Join: @IITJEE_Advanced

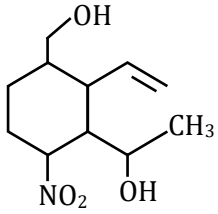
9. The major product obtained in the following reaction is :- **[JEE (Main) 2019]**



(1) 

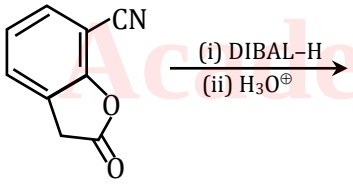
(2) 

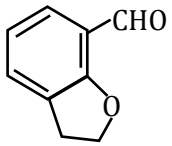
(3) 

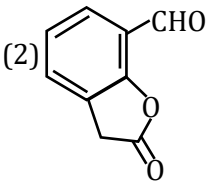
(4) 

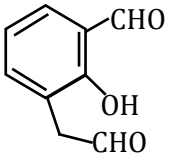
CRD038

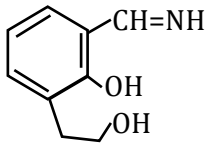
10. The major product of the following reaction is: **[JEE (Main) 2019]**



(1) 

(2) 

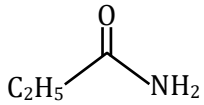
(3) 

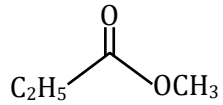
(4) 

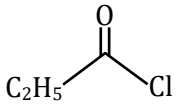
CRD039

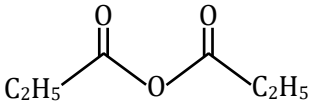
11. The increasing order of the reactivity of the following with $LiAlH_4$ is :

[JEE (Main) 2019]

(A) 

(B) 

(C) 

(D) 

(1) (A) < (B) < (D) < (C)

(2) (A) < (B) < (C) < (D)

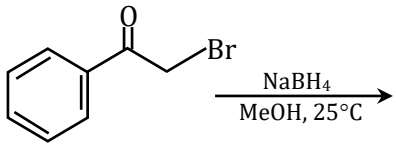
(3) (B) < (A) < (D) < (C)

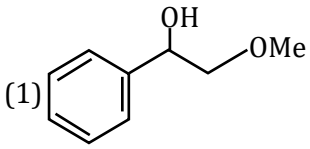
(4) (B) < (A) < (C) < (D)

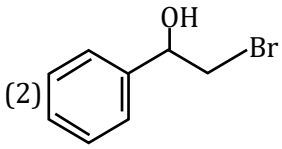
CRD040

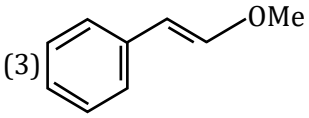
For More Join: @IITJEE_Advanced

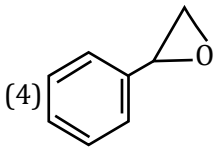
12. The major product of the following reaction is : **[JEE (Main) 2019]**



(1) 

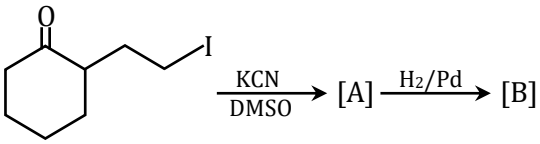
(2) 

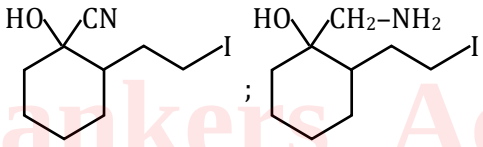
(3) 

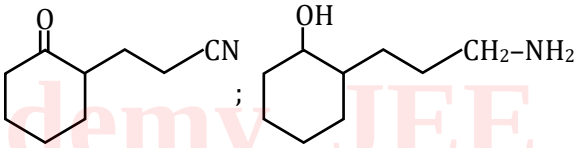
(4) 

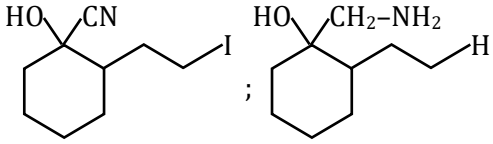
CRD041

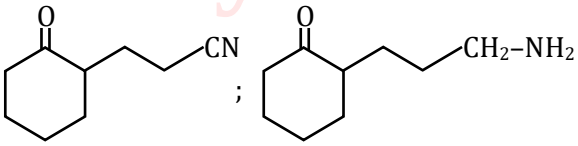
13. The major products A and B for the following reactions are, respectively: **[JEE (Main) 2019]**



(1) 

(2) 

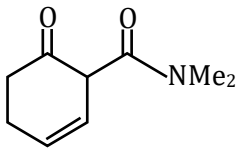
(3) 

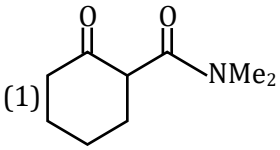
(4) 

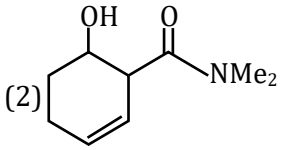
Rankers' Academy JEE

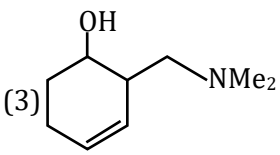
CRD042

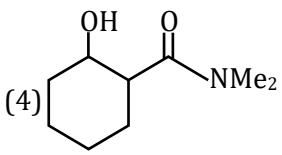
14. The main reduction product of the following compound with $NaBH_4$ in methanol is :- **[JEE (Main) 2018]**



(1) 

(2) 

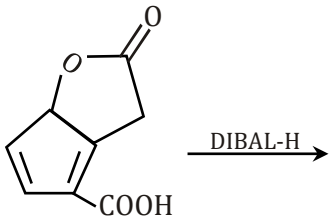
(3) 

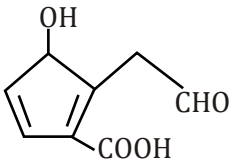
(4) 

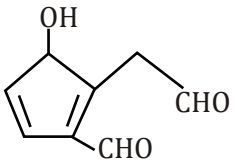
CRD034

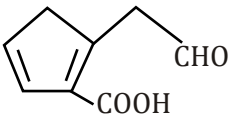
For More Join: @IITJEE_Advanced

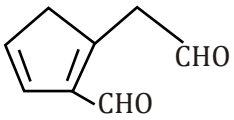
15. The major product obtained in the following reaction is : **[JEE (Main) 2017]**



(1) 

(2) 

(3) 

(4) 

CRD031

16. Wilkinson catalyst is : **[JEE (Main) 2019]**

(1) $[(Ph_3P)_3RhCl]$ $(Et = C_2H_5)$ (2) $[Et_3P)_3IrCl]$

(3) $[Et_3P)_3RhCl]$ (4) $[Ph_3P)_3IrCl]$

CRD036

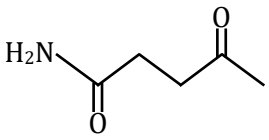
17. The trans-alkenes are formed by the reduction of alkynes with : **[JEE (Main) 2018]**

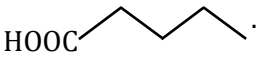
(1) $NaBH_4$ (2) Na/liq.NH_3 (3) Sn–HCl (4) H_2–Pd/C, $BaSO_4$

CRD033

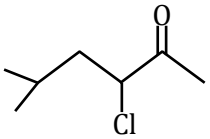
Rankers Academy JEE

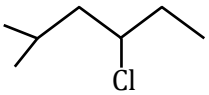
18. Given below are two statements : **[JEE (Main) 2023]**

Statement I : 

under Clemmensen reduction conditions will give 

.

Statement II : 

under Wolff-Kishner reduction condition will give 

In the light of the above statements, choose the correct answer from the options given below :

(1) Statement I is false, but Statement II is true

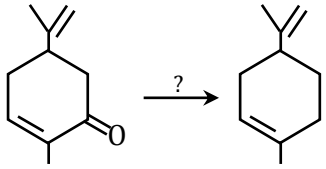
(2) Both Statement I and Statement II are false

(3) Statement I is true, but Statement II is false

(4) Both Statement I and Statement II are true

CRD050

For More Join: @IITJEE_Advanced

19.  **[JEE (Main) 2021]**

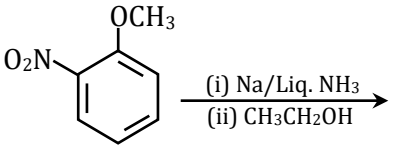
Which of the following reagent is suitable for the preparation of the product in the above reaction?

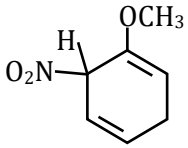
(1) $NaBH_4$ (2) $NH_2–NH_2/C_2H_5O^{\ominus}Na^{\oplus}$

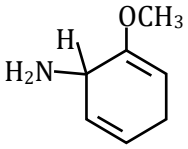
(3) Ni/H_2 (4) Red P + Cl_2

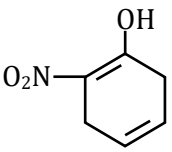
CRD046

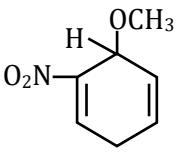
20. The major product of the following reaction is **[JEE (Main) 2022]**



(1) 

(2) 

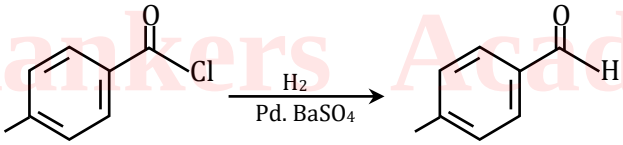
(3) 

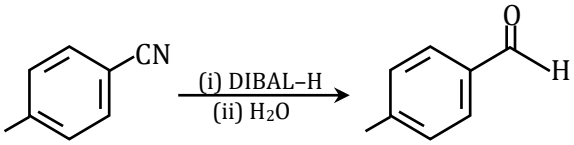
(4) 

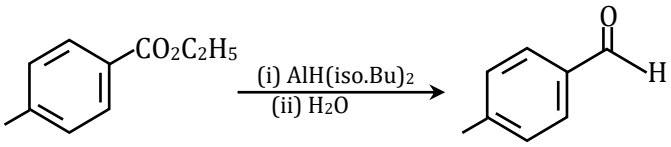
CRD047

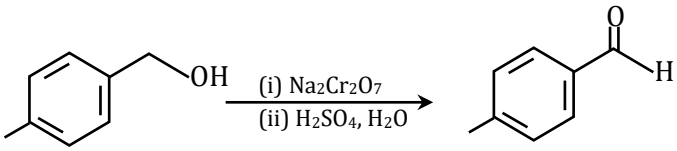
21. Which one of the following reactions does not represent correct combination of substrate and product under the given conditions ? **[JEE (Main) 2022]**

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(1) 

(2) 

(3) 

(4) 

CRD048

22. Glucose on prolonged heating with HI gives : **[JEE (Main) 2018]**

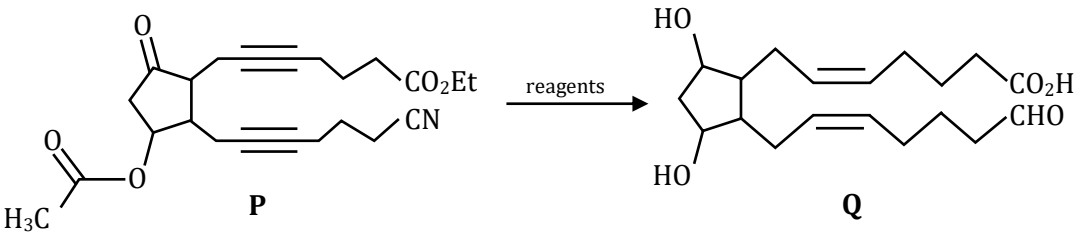
(1) 1–Hexene (2) Hexanoic acid (3) 6-iodohexanal (4) n-Hexane

CRD032

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EXERCISE - JEE (Advanced) PYQ

1. The option(s) with correct sequence of reagents for the conversion of **P** to **Q** is(are) : **[JEE (Advanced) 2024]**



P **Q**

(A) i) Lindlar's catalyst, H_2 ; ii) $SnCl_2/HCl$; iii) $NaBH_4$; iv) H_3O^+
(B) i) Lindlar's catalyst, H_2 ; ii) H_3O^+ ; iii) $SnCl_2/HCl$; iv) $NaBH_4$
(C) i) $NaBH_4$; ii) $SnCl_2/HCl$; iii) H_3O^+ ; iv) Lindlar's catalyst, H_2
(D) i) Lindlar's catalyst, H_2 ; ii) $NaBH_4$; iii) $SnCl_2/HCl$; iv) H_3O^+

CRD134

2. Consider all possible isomeric ketones including stereoisomers of MW = 100, All these isomers are independently reacted with $NaBH_4$ (**NOTE :** stereoisomers are also reacted separately). The total number of ketones that give a racemic product(s) is/are. **[JEE (Advanced) 2014]**

CRD081

3. In the reaction,

$CH_3COOH \xrightarrow{LiAlH_4} A \xrightarrow{PCl_5} B \xrightarrow{AlC.KOH} C$, Final product C is. **[JEE (Advanced) 2014]**

(A) Ethylene (B) Acetyl chloride (C) Acetaldehyde (D) Acetylene

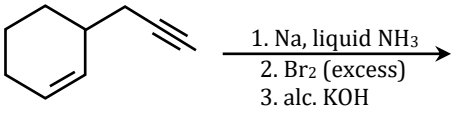
CRD082

4. When $CH_2=CH-COOH$ is reduced with $LiAlH_4$, the compound obtained will be - **[AIEEE 2003]**

(A) $CH_3-CH_2-CH_2OH$ (B) CH_3-CH_2-CHO (C) CH_3-CH_2-COOH (D) $CH_2=CH-CH_2OH$

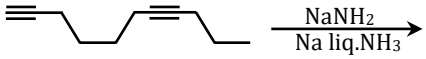
CRD076

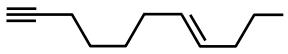
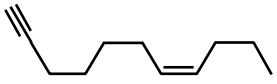
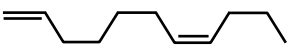
5. The number of isomeric tetraenes (NOT containing sp-hybridized carbon atoms) that can be formed from the following reaction sequence is _____. **[JEE (Advanced) 2022]**



CRD085

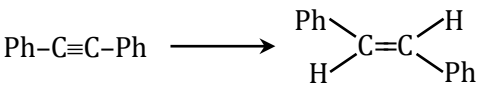
6. The major product formed in the following reaction is **[JEE (Advanced) 2021]**



(A) 
(B) 
(C) 
(D) 

CRD084

7. The reagent needed for converting is : **[JEE (Advanced) 2014]**



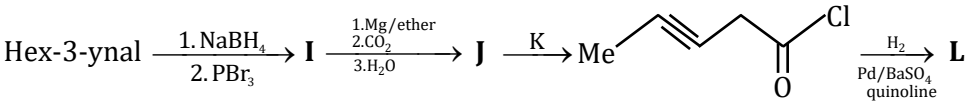
(A) H_2 / Lindlar Cat. (B) Cat. hydrogenation
(C) $LiAlH_4$ (D) Li / NH_3

CRD083

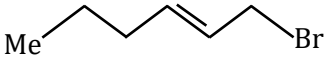
For More Join: @IITJEE_Advanced

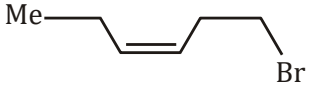
Paragraph for Question No. 08 to 10

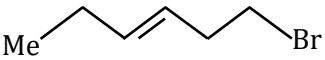
In the following reaction sequence, products **I**, **J** and **L** are formed. **K** represents a reagent.

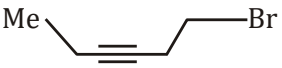


8. The structure of the product **I** is **[JEE (Advanced) 2008]**

(A) 

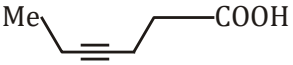
(B) 

(C) 

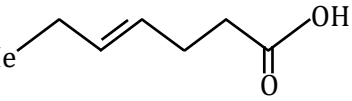
(D) 

CRD078

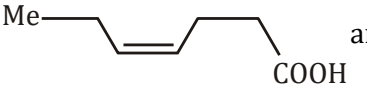
9. The structures of compounds **J** and **K**, respectively, are **[JEE (Advanced) 2008]**

(A) 

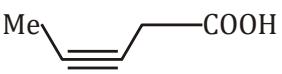
and $SOCl_2$

(B) 

and SO_2Cl_2

(C) 

and $SOCl_2$

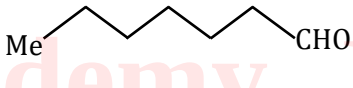
(D) 

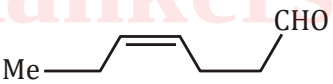
and CH_3SO_2Cl

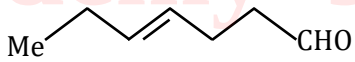
CRD079

10. The structure of product **L** is : **[JEE (Advanced) 2008]**

(A) 

(B) 

(C) 

(D) 

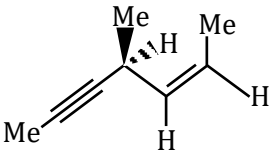
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CRD080

11. 2–hexyne can be converted into trans–2–hexene by the action of : **[IIT 2004]**

(A) H_2–Pd-$BaSO_4$ (B) Li in liq. NH_3 (C) H_2–PtO_2 (D) $NaBH_4$

CRD077

12. 

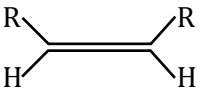
Hydrogenation of the above compound in the presence of poisoned palladium catalyst gives- **[IIT 2001]**

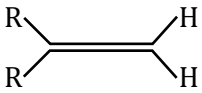
(A) An optically active compound (B) An optically inactive compound

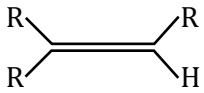
(C) A racemic mixture (D) A diastereomeric mixture

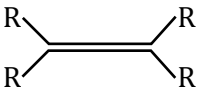
CRD075

13. Which one of the following alkenes will react fastest with H_2 under catalytic hydrogenation condition **[IIT 2000]**

(A) 

(B) 

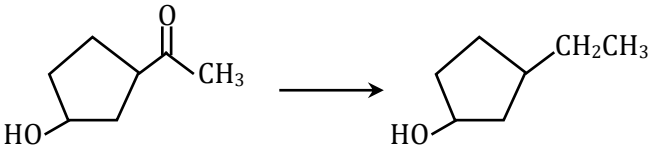
(C) 

(D) 

CRD073

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14. The appropriate reagent for the following transformation: **[IIT 2000]**



(A) Zn (Hg), HCl (B) NH_2NH_2, $OH^{\ominus}$ (C) H_2/Ni (D) $NaBH_4$

CRD074

15. Under Wolff Kishner reduction conditions, the conversions which may be brought about is? **[IIT 1995]**

(A) Benzaldehyde into Benzyl alcohol (B) Cyclohexanol into Cyclohexane
(C) Cyclohexanone into Cyclohexanol (D) Benzophenone into Diphenylmethane

CRD072

16. Hydrogenation of benzoyl chloride in the presence of Pd and $BaSO_4$ gives : **[IIT 1992]**

(A) Benzyl alcohol (B) Benzaldehyde (C) Benzoic acid (D) Phenol

CRD071

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ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	A	D	B	A	C	D	D	A

EXERCISE (O-2)

Que.	1	2	3	4	5	6
Ans.	ABCD	ABC	ABC	ABC	ABD	C

EXERCISE (S)

Que.	1	2	3	4	5
Ans.	5	3	3	1.06	4.00

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	1	3	1	3	3	4	4	2	3
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	1	4	2	2	1	1	2	3	2	1
Que.	21	22								
Ans.	4	4								

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EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	CD	5	A	D	2	B	D	D	A	C
Que.	11	12	13	14	15	16				
Ans.	B	B	A	B	D	B				

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